



CS 584 Course Project Guidelines

Spring 2026

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CS584 Final Course Project

Grouping:

Team: 3 students form a team (total we have 12-15 teams)
Submit team sign-up contract by 2/5

Deliverables:

Proposal presentation on 2/12, submit slides
Mid-stage presentation on 4/16, submit slides
Final presentation on 5/5, submit poster
Final report due on 5/14

Guidelines:

We will provide presentation templates (.ppt) and report template (in latex).



Potential Project Topics

Need to assess scope and feasibility

- **Topic 1: Learning techniques for text**
 - Investigate the advantages/disadvantages of different techniques for processing and understanding text
- **Topic 2: Bias and fairness**
 - Investigate bias, fairness, data imbalance in NLP applications
- **Topic 3: Privacy and security in NLP**
 - Explore privacy leakage and data poisoning attacks/defense in NLP techniques
- **Topic 4: Efficient NLP**
 - Explore techniques to improve pre-training/fine-tuning, distributed training in NLP
- **Topic 5: Language models for code or math proofs**
 - How rigorous are LLMs for code generation or theoretical proofs?



Potential Project Topics

Need to assess scope and feasibility

- **Topic 6: Agentic AI**

- Consider a type of problems where an agent needs to first make a plan and then proceed with the items on the plan. Test how well an AI agent can make a plan that is adequate for its own capability. Software to consider: [AgentBench](#)

- **Topic 7: Intervention of reasoning**

- Chain-of-thought tampering: Edit a few randomly/strategically/algorithmically selected words. See how well the model's output can be changed.
- Multi-turn intervention. At every conversation iteration (or at fixed intervals), insert a strategically/algorithmically selected sequence of words. See if the model's math reasoning capability can be improved. Dataset: e.g., Math-500

- **Topic 8: Prompt steering foundations**

- Given a few examples of steered results, recover the prompt that was used to steer the language model. Related papers: Morris et al (2023) [LM inversion](#) or [Gao et al \(2024\) Deliberate prompt recovery](#)
- Given an LLM and a steering goal, study whether it is possible to steer the LLM towards that goal. Related papers: [Wang et al \(2025\) Distributional Prompting](#) or [Miehling et al \(2025\) Prompt Steerability](#)

Potential Project Topics (more challenging)

Need to assess scope and feasibility

- **Topic 9: Awareness of context engineering**

- When the contexts become long, the algorithms may compress the contexts, or replace them with only filenames. Test whether an AI agent knows the context is compressed. Dataset to consider: a RAG system with file structures. A related idea: tamper awareness. Sometimes the files are modified unexpectedly. Test whether the AI agent knows the files are modified.

- **Topic 10: Multi-agent orchestration**

- Improve the task performance: try change the roles of each of the agents, or change the way the messages are passed between these agents. Tool to consider: [LangGraph](#)
- Improve the memory: try compressing the context.
- Improve the collaboration graph: Their interactions form a graph (e.g., each node is an “execution”, and each edge is a “delegation”). Optimize this graph, so that the overall collaboration is made more efficient.

- **Topic 11: Mechanistic interpretability for safety**

- Using {representation probes, inserted natural language query}, assess if the model is aware of the final answer halfway during reasoning. Dataset: Math-500
- Establish {representation probes, inserted natural language query} assess if the model knows if it is lying. Dataset: TruthfulQA

- **Topic 12: Choose your own**

1- Learning techniques for text

Advantages and limitations of different techniques for processing and understanding text

- **Core Questions**
 - What assumptions do different text learning techniques make about language structure?
 - How do these techniques trade off **performance, interpretability, robustness, and efficiency**?
 - Under what data regimes (low-resource vs large-scale) do they succeed or fail?
- **Project Directions**
 - Compare **statistical methods** (TF-IDF, n-grams, topic models) with **neural approaches** (RNNs, CNNs, Transformers).
 - Study **representation learning**: static embeddings (word2vec, GloVe) vs contextual embeddings (BERT, GPT).
 - Analyze **failure modes**, such as sensitivity to syntax, word order, or domain shift.
- **Deliverables**
 - Experimental comparison with clear metrics
 - Error analysis with qualitative examples
 - A discussion of when “simpler” methods may be preferable



2- Bias and fairness

Bias, fairness, and data imbalance in NLP systems

- **Core Questions**
 - What types of bias (demographic, societal, label imbalance) arise in NLP?
 - How do biases propagate from data → representations → predictions?
 - What does “fairness” mean in the context of language?
- **Project Directions**
 - Measure bias in embeddings, classifiers, or generative models.
 - Analyze data imbalance and its impact on minority groups.
 - Compare bias mitigation strategies.
- **Deliverables**
 - Quantitative bias evaluation
 - Visualization of bias before/after mitigation
 - Discussion of trade-offs between fairness and performance

Caliskan, A., Bryson, J. J., & Narayanan, A. (2017). Semantics derived automatically from language corpora contain human-like biases. *Science*.

Blodgett, S. L. et al. (2020). Language (Technology) is Power: A Critical Survey of Bias in NLP. *ACL*.

3- Privacy and Security in NLP

Privacy leakage and adversarial attacks/defenses in NLP models

- **Theme: Core Questions**
 - How can NLP models leak sensitive training data?
 - How vulnerable are models to poisoning or adversarial manipulation?
 - What defenses are effective, and at what cost?
- **Project Directions**
 - Membership inference or data extraction attacks on language models
 - Data poisoning attacks in text classification or pretraining
 - Privacy-preserving training techniques.
- **Deliverables**
 - Attack success metrics
 - Defense effectiveness analysis
 - Clear threat model and assumptions

Carlini, N. et al. (2021). Extracting Training Data from Large Language Models. USENIX Security.

Jagielski, M. et al. (2018). Manipulating Machine Learning: Poisoning Attacks and Countermeasures. IEEE S&P.

4- Efficient NLP

Improving efficiency in training and deploying NLP models

- **Core Questions**
 - Where are the major computational bottlenecks in NLP?
 - How do efficiency methods affect accuracy and generalization?
 - Can we reduce cost without sacrificing performance?
- **Project Directions**
 - Parameter-efficient fine-tuning (LoRA, adapters, prefix tuning)
 - Model compression (pruning, quantization, distillation)
 - Distributed or parallel training strategies.
- **Deliverables**
 - Performance–efficiency trade-off plots
 - Reproducible experimental setup
 - Discussion of scalability implications

Hu, E. et al. (2022). LoRA: Low-Rank Adaptation of Large Language Models. ICLR.

Sanh, V. et al. (2019). DistilBERT: A Distilled Version of BERT. NeurIPS Workshop.

5- Language models for code or math proofs

Rigor and reliability of LLMs in formal reasoning

- **Core Questions**
 - Where do LLMs succeed or fail in logical rigor?
 - Do models hallucinate plausible but incorrect code or proofs?
 - Can verification or constraints improve reliability?
- **Project Directions**
 - Evaluate code generation correctness beyond unit tests.
 - Test mathematical reasoning on structured proof tasks.
 - Compare chain-of-thought vs direct answers..
- **Deliverables**
 - Error taxonomy of failures
 - Quantitative rigor metrics
 - Insights on when LLMs can be trusted

Chen, M. et al. (2021). Evaluating Large Language Models Trained on Code.

Lewkowycz, A. et al. (2022). Solving Quantitative Reasoning Problems with Language Models. NeurIPS.

6- Agentic AI

Planning and execution in AI agents.

- **Core Questions**
 - Can an agent generate plans aligned with its own capabilities?
 - How does planning quality affect downstream success?
 - Where do agents fail: planning or execution?
- **Project Directions**
 - Evaluate agents in multi-step tasks using AgentBench.
 - Compare planning-first vs reactive agents.
 - Study plan feasibility and adaptability.
- **Deliverables**
 - Plan quality metrics
 - Failure case analysis
 - Discussion of self-awareness and capability estimation

Liu, X. et al. (2023). AgentBench: Evaluating LLMs as Agents. arXiv.

Yao, S. et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR.

7- Intervention of Reasoning

Manipulating and improving LLM reasoning through interventions.

- **Core Questions**
 - How fragile is chain-of-thought reasoning?
 - Can targeted interventions improve reasoning reliability?
 - Are improvements task-specific or general?
- **Project Directions**
 - Chain-of-thought tampering
 - Random vs strategic edits
 - Measure output sensitivity
 - Multi-turn interventions
 - Insert guiding tokens/prompts periodically
 - Evaluate improvement on math reasoning.
- **Deliverables**
 - Sensitivity analysis plots
 - Comparison of intervention strategies
 - Theoretical discussion of why interventions work

Wei, J. et al. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS.

Turpin, M. et al. (2024). Language Models Don't Always Say What They Think: Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting.

8- Prompt Steering Foundations

Understanding and controlling prompt-based steering.

- **Core Questions**
 - Is prompt steering recoverable or identifiable?
 - What are the limits of steering an LLM?
 - How stable is steering across tasks and models?
- **Project Directions**
 - Prompt recovery: Given steered outputs, infer the hidden prompt
 - Prompt steerability: Test whether a goal can be consistently induced
- **Deliverables**
 - Recovery accuracy or steering success rates
 - Failure modes and impossibility cases
 - Conceptual insights into prompt-based control

Morris, J. et al. (2023). Language Model Inversion. arXiv.

Gao, L. et al. (2024). DORY: Deliberative Prompt Recovery for LLM.

Wang, X. et al. (2025). Distributional Prompting. arXiv.

Bad ideas for a course project

- ✗ Build a large language model from scratch
- ✗ Evaluate 100 LLMs on 100 Large datasets

Grade Distribution

- Team project (35%)
 - 5% Proposal presentation in class
 - 10% Mid-stage presentation in class
 - 10% Final poster presentation → One-day symposium
 - 10% Final project report (Use LaTeX, template will be provided)
- Presentation judge
 - Dr. Ning and TA
 - Invited judge (TBD)
 - From industry, from cross-disciplinary

Proposal Presentation Outline (each team has 7 mins)

- Introduce the topic you will work on
 - What is this topic
 - Why choose this topic
 - How important the topic is
- What do you plan to do under the topic?
 - Introduce your idea
 - Please make sure the idea is feasible given the amount of time
 - Please make sure the idea is well justified and reasonable
- A plan to achieve this goal
 - Milestones, step-wise design, ...
 - It would be fine if your plan changed a little bit at the final stage
- Expected timeline and milestone
 - By which date, which major feature will be completed
 - How you distributed the workloads



Mid-stage Presentation Outline (each team has 7 mins)

- Introduce the topic you will work on
 - What is this topic
 - How important the topic is
- What have you done under the topic?
 - Introduce your idea
 - Progress
 - Method design
 - Experimental design and evaluation
 - Results
- Expected poster outcomes:
 - What will be included in the poster
 - What will be included in the final report



Grading Rubric: Presentation

Category	low	medium	high
Content and Knowledge	Lack of sufficient technical content and sophistication is evident in several components of the project.	Project analyzes a sufficient amount of pertinent technical information in the development of a few feasible solutions to meet the project objective. Relate to what we have learned.	Project analyzes a diverse and significant amount of pertinent technical information in the development of several feasible solutions to meet the project objective. Project also incorporates and analyzes possible solutions that seem counter-intuitive, surprising, or contrary to conventional approaches. Relate to what we have learned.
Critical Thinking	Fails to present critical thinking and has no opinions about the content presented	Begins to relate alternative views. Rough integration of multiple viewpoints. Ideas are investigated in a limited way. May overstate conflict or dismiss alternative views hastily. Analysis of other views mostly accurate. Some evidence of self-assessment.	Addresses diverse perspectives from a variety of sources to qualify analysis. Any analogies are used effectively. Clearly justifies own view while respecting views of others. Analysis of other positions is accurate and respectful. Evidence of reflection and self-assessment.
Organization & Presentation	Presenter did not give a clear and concise introduction of the topic. The flow of the presentation was unclear and confusing to follow. The presenter was un-enthusiastic; the audience's attention showed disinterest and the purpose for the presentation was not achieved.	Presenter gave a clear and concise introduction of the topic. The flow of the presentation was clear and concise and easy to follow. The presenter kept the attention of the audience the whole time and the purpose of the presentation was achieved.	The introduction and actual presentation were exceptionally well- organized and easy to understand. The presenter sustained the interest of the audience in clever and innovative ways and achieved the purpose of the presentation.
Questions and Answers	Speaker is unable to answer questions; loses control of the process	Speaker has some difficulty answering questions concisely; some problems responding to some questions (e.g., hostile questions, aggressive questions)	Speaker answers questions knowledgeably, thoroughly, and concisely; process is handled smoothly



Grading Rubric: Final Report

Category	low	medium	high
Literature Review	The report introduced and discussed few related work	The report introduced and discussed related work	The report introduced and discussed related work, particularly gave detailed description of the state-of-the-art and the most related studies. Besides, the report can distinguish with existing studies
Contribution	The report has presented few unique contribution and discussion	The report has presented some unique contribution and thoughtful discussion	The report has shown substantial unique contribution and thoughtful discussion
Organization & Presentation	There is a few writing issues, such as typos or grammar mistakes. The report does not have a clear logic flow, and the report is hard to follow. Less than 5 pages (excluding the references).	The report is readable and present some helpful visualizations to improve readability The report is organized with a clear logic flow, it's easy to follow. Well-formated references	The report is well written with intuitive and elegant figures, tables, and data visualization. There is few writing issues, such as typos or grammar mistakes. The report is organized with a clear logic flow and well-defined components. The objective, methodology, and evaluation of this report are consistent. Well-formated references

One-day AI symposium (tentatively scheduled for 5/5)

Morning

- Keynote talks by invited speakers (from industry)

- Research presentations

Afternoon

- Final project poster presentation & grading

